STRESS MANAGEMENT WIRE & HOOP GAME - USER GUIDE

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The Stress Management Wire & Hoop Game is an enhanced version of the classic wire & hoop game. This version of the traditonal game introduces stress-inducing components, including a vibrating handle and red flashing lights.

The project is pioneered by SDG 3, so the game's main idea is to promote practical stress management techniques such as taking control and taking a break to maintain good mental well-being.

REQUIRED ELEMENTS (INCLUDED IN BOX)

- NXP FRDM-MCXN947 Microcontroller and connecting cable
- Black plastic base casing
- Wooden mounting stand for track
- Wire hoop
- Regular wires, and jumper wires
- USB Plug

SET UP INSTRUCTIONS

STEP ONE - GATHER ALL COMPONENTS

The first step is to gather all the components listed above. The microcontroller will come flashed with the required code, and will be good to run from the box.

STEP TWO - POWER THE MICROCONTROLLER

After verifying all components are present, you should power the microcontroller using the white USB plug provided.

STEP THREE - PLAY THE GAME!

Next, wait for 15 seconds for the microcontroller to calibrate and the OLED screen should display some text confirming the game is ready to setup.

As shown on the OLED screen, place the hoop on the silver starting plate until the Green LEDs begin flashing.

To play the game, place your the pointer finger of your least dominant hand on the heart rate monitor. Once you have lifted the hoop from the silver starting plates the timer will start.

On the wire track there is a silver section - this is a rest area. At this point the timer will pause, and contact with the wire will not deduct points from your score.

There are two hoop sizes available, with varying difficulties, so if the game seems too difficult, please do change the hoop size.

As for scoring, the number of contacts with the wire and hoop is deducted from a starting score of 100. At the end of the game, the players' final score is displayed alongside the time taken to complete it.

Return the hoop to the starting plate to restart the game.

TROUBLESHOOTING

The game is designed to run through a loop, when an error is detected the loop stops and the error is displayed on the OLED screen.

The two potential errors are: 'MAX ... 903' and MAX ... 900'.

'MAX ... 903' means that the board could not read or write data properly, and 'MAX ... 900' means that the board could not initialise properly. The solutions to both of these problems are to simply unplug the microcontroller and plug it back in.

In the case of the OLED screen not displaying anything on start up, either try the above method, or disconnect the OLED board and follow the instructions below to read the DEBUG outputs.

READ THE DEBUG OUTPUT

Plug the USB into any laptop / PC, and open the MCUXpresso IDE. Select the terminal tab, select open a new terminal, then input the options below and select 'OK':

Choose Terminal: Serial Terminal
Serial Port: COM3 (your number may vary based on which USB port is connected)
Encoding: Default (ISO-8859-1)

The output from here will be the same as if the OLED was connected and working.

Another option for debugging the system is checking the pins are connected correctly and securely. Reference the following tables when checking connections between the microcontroller and the rest of the system:

SIGNAL DESCRIPTION	PIN NO.	WIRING DESTINATION
VOLTAGE SUPPLY (3V3)	P3V3 | J8_1	MAX - V_{IN} (red)
GROUND SUPPLY (GND)	GND | J8_2	MAX – GND (black)
I_2C SDA	P4_0 | J8_4	MAX - SDA (pink)
I_2C SCL	P4_1 | J8_3	MAX - SCL (blue)
SAMPLE INTERRUPT	P5_8 | J9_31	MAX - INT (gray)

MAXREFDES117# WIRING DESCRIPTION

SIGNAL DESCRIPTION	PIN NO.	WIRING DESTINATION
START DETECT	P1_12 | J9_28	Contact at start of Track
TRACK TOUCH DETECT	P1_23 | J9_23	Wire Track
BREAK DETECT	P1_7 | J9_9	Wire Wrap on Track
FINISH DETECT	P1_22 | J9_24	Contact at end of Track

GAME LOGIC GPIO WIRING DESCRIPTION

SIGNAL DESCRIPTION	PIN NO.	WIRING DESTINATION
MOTOR DRIVER PWM	P2_4 | J3_11	PCB – Motor Supply PWM
LED DRIVER PWM	P2_6 | J3_15	PCB – LED Supply PWM
LED GREEN SELECT	P5_9 | J9_29	PCB – GREEN LED Select
LED RED SELECT	P2_0 | J9_25	PCB – RED LED Select
SPEAKER CONTROL	P1_23 | J9_23*	PCB – Speaker Control

PWM OUTPUTS WIRING DESCRIPTION

SIGNAL DESCRIPTION	PIN NO.	WIRING DESTINATION
VOLTAGE SUPPLY (5V)	P5V0 | J3_10	PCB - +5V Supply
GROUND SUPPLY (GND)	GND | J6_8	PCB – GND Supply

PCB VOLTAGE SUPPLY WIRING DESCRIPTION

PIN NO.	WIRING DESTINATION
P4_1 | J2_20	Shield – J2_20
P4_0 | J2_18	Shield – J2_18
VDDA | J2_16	Shield – J2_16
GND | J2_14	Shield – J2_14
P3V3 | J3_4	Shield – J3_4
GND | J3_12	Shield – J3_12
GND | J2_14	Shield – J3_14

OLED SHIELD RISER WIRING DESCRIPTION

DEMONSTRATING TO AN AUIDENCE

When running a demonstration to a group, there are a few important notes:

- It's important to give a brief overview of the vision behind the game and discuss the link to SDG 3. This is outlined in the introduction of the user manual.
- During the setup, the demonstrator should discuss the rules of the game, which are detailed in step four.
- After gameplay is confirmed, the game is ready to play!
- The demonstrator should discuss how the game is making them feel and how their stress levels are increasing, even from having to multitask, as well as how variations to the game would increase / decrease their stress.

PACK UP INSTRUCTIONS

To pack up the game, the player should first unplug the microcontroller for safety purposes, and then detach the wire track from the base with a gentle pull.

After that, place the black base casing into the plastic box, and lay the wire track flat on top of the base.